Checker and diagnostics

These files are written every run, regardless of `print.prt`. Check them first when validating a simulation.

simulation.out

Source

Opened in src/main.f90 on unit 9003 (first 30 lines). Daily progress lines written from src/time_control.f90. Climate-file open events written from src/proc_date_time.f90.

Content

- Banner: SWAT+ version, revision, compiler, OS, build date.
- One line per climate file as it is opened.
- One line per simulated day:

  `<scenario name>  <mon>  <day>  <year>  Yr  <yr>  of  <yr_total>  Time  hh:mm:ss`
- On successful completion, a line `Execution successfully completed` and a `Date of Sim` stamp.

Example

```
                  SWAT+
             Revision 61.0.1-33-g68b1328
      Soil & Water Assessment Tool
IntelLLVM (2024.0.0), 2024-08-27 08:49:23, Linux
    Program reading . . . executing


  Date of Sim   8/27/2024 Time   8:49:51
 reading from pet file                  Time   8:49:51
 reading from precipitation file        Time   8:49:51
 ...
  Original Simulation        1   1  1975 Yr    1 of   46 Time   8:49:51
  ...
  Original Simulation       12  31  2020 Yr   46 of   46 Time   8:49:53

 Execution successfully completed

  Date of Sim   8/27/2024 Time   8:49:53
```

If `Execution successfully completed` is absent, the run aborted. Search upward in the file for the last day printed: the failure is on the next simulated day.

diagnostics.out

Source

Opened in src/proc_bsn.f90 on unit 9001. Lines are appended by every `*_read.f90` that detects a missing input reference (`grep -rn "write *(9001"` lists about 60 call sites in the source).

Content

One warning per line. Each warning is the name of a missing reference and the input file that pointed to it. Example:

```
DIAGNOSTICS.OUT FILE
corn_lum                                CORN_rot                                not found in management.
agrr_lum                                AGRR_rot                                not found in management.
soyb_lum                                SOYB_rot                                not found in management.
ames.tem                                        file not found (tgage)
file not found (basins_carbon.tes)
```

Read this file end-to-end after every run. Lines here mean SWAT+ silently substituted a default or skipped a setting because the referenced object was not in the database it was supposed to be in. Common categories:

- `<name> not found in management.sch`. A land use management points at a schedule that does not exist.
- `<name> not found (initial.cha)`, `(hydrology.cha)`, `(sediment.cha)`, `(nutrients.cha)`. A channel object points to a property set that does not exist.
- `file not found (<gage>)`. A weather station refers to a gage file that is missing.
- `file not found (<file>)`. A control input (carbon coefficients, snow database) is missing.

An empty `diagnostics.out` (only the header) means no missing references were detected.

checker.out

Source

Opened in src/proc_hru.f90 on unit 4000. Writes one row per HRU after model setup. Header types are `output_checker_header` and `output_checker_unit` in src/hydrograph_module.f90.

Format

- Line 1: scenario name and SWAT+ version.
- Line 2: column header (`chk_hdr`).
- Line 3: unit row (`chk_unit`). The last cell carries the legend `0=notile;1=tile;`.
- Lines 4+: one row per HRU.

#	Column	Units	Description
1	sname	name	soil name
2	hydgrp	A/B/C/D	hydrologic soil group
3	zmx	mm	maximum rooting depth (`soil%zmx`)
4	usle_k	none	USLE soil erodibility factor
5	sumfc	mm	sum of field capacity over profile
6	sumul	mm	sum of saturation (upper limit) over profile
7	usle_p	none	USLE practice factor
8	usle_ls	none	USLE slope-length-steepness factor
9	esco	none	soil evaporation compensation factor
10	epco	none	plant uptake compensation factor
11	cn3_swf	none	curve-number soil-water adjustment
12	perco	none	percolation coefficient
13	latq_co	none	lateral flow coefficient
14	tiledrain	0 or 1	tile drainage flag

Example

`refdata/Ames_sub1/checker.out`:

```
 demo                      SWAT+ 2024-08-27        MODULAR Rev 2024.61.0.1-33-g68b1328
                                        zmx       usle_k       sumfc       sumul      usle_p     usle_ls
 sname            hydgrp                (mm)                    (mm)        (mm)
soil_01           B                2000.0000       0.0000     361.5000    616.1781      1.0000      0.3638
soil_02           B                2000.0000       0.0000     361.5000    445.6733      1.0000      0.3638
```

Check `usle_k`, `zmx`, `sumfc`, and `sumul` for unrealistic values (zero, negative, or extreme). A `zmx` of 2000 mm on every HRU usually means the soil definitions were not parsed and the model fell back to a default.

files_out.out

Source

Opened in src/proc_bsn.f90 on unit 9000. Every `open_output_file` call that opens an output writes one line here, naming the file and a short tag.

Format

- Line 1: header `files_out.out - OUTPUT FILES WRITTEN`.
- Lines 2+: two columns, tab/space separated.

Column	Description
tag	`CHK`, `HRU`, `BASIN`, `CHN`, `AQU`, `RES`, `MGT`, …
filename	the file SWAT+ opened

This is the master inventory: anything actually produced by this run is listed here. If you expect a file and it is missing from `files_out.out`, the corresponding switch in `print.prt` was `n`.

erosion.out

Source

Opened in src/proc_hru.f90 on unit 4001. Writes one row per HRU of the USLE inputs and computed factors. Always on; not switchable in `print.prt`.

success.fin

Created in src/main.f90 on unit 107. An empty file written only at the end of a successful run. External tools (calibration drivers, test harnesses) test for its existence to confirm the run completed.

erosion.txt

Opened in `main.f90` on unit 888. Internal debug log of erosion calculations. Not part of the documented output set.

Recommended checks after every run

1. Open `simulation.out` and confirm the last line is `Execution successfully completed`. If not, look at the last printed day.
2. Open `diagnostics.out`. Every line is a problem. Resolve each one before trusting any output.
3. Open `checker.out`. Confirm `zmx`, `sumfc`, `sumul` are sane per HRU. Confirm `tiledrain` flag matches your design.
4. Confirm `success.fin` exists.
5. Open `files_out.out`. Confirm every output file your `print.prt` requested is listed.

Related files

- Naming convention.
- Post-processing helpers. Includes a parser for `checker.out`.